



Introduction to Cyber security

Data is gold, Save it like a treasure

Course by MS. Misbah Aziz



What we cover

UNIT -I Introduction to Cyber Security:

- Basic Cyber Security Concepts
- Challenges and Constraints in cyber security
- Vulnerabilities, threats and attacks
- Cyber Crime VS Cyber security
- CIA Triad (confidentiality ,integrity and availability)
- Types of attacks (DOS,DDOS,PHISHING,MALVERTISING,ETC)



What cyber security is???

Cyber security refers to the body of technologies, processes, and practices designed to protect networks, devices, programs, and data from attack, damage, or unauthorized access. Cyber security may also be referred to as information technology security.



According to S. 1901 “Cybersecurity Research and Education Act of 2002”

cybersecurity: “information assurance, including scientific, technical, management, or any other relevant disciplines required to ensure computer and network security, including, but not limited to, a discipline related to the following functions: operations.

(A) Secure System and network administration and operation.

(B) Systems security engineering.

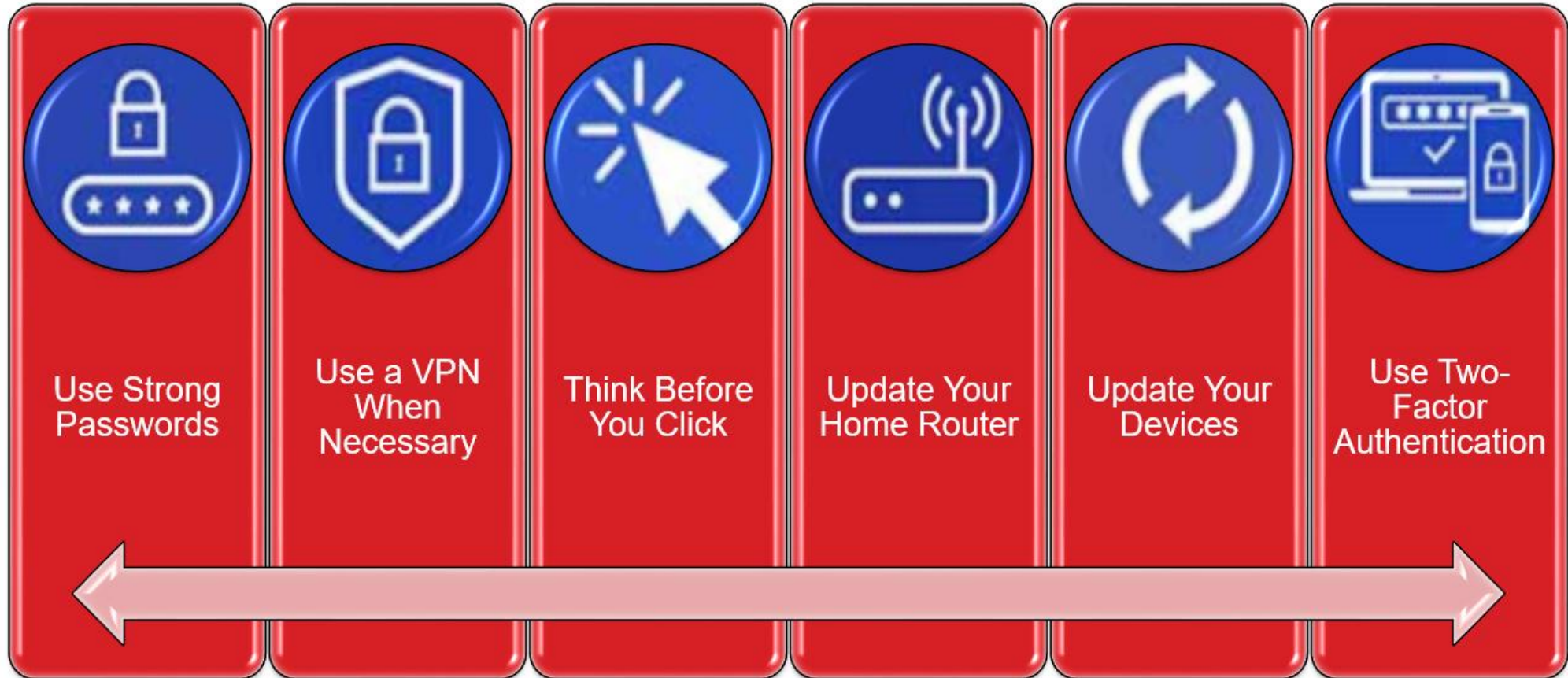
(C) Information assurance systems and product acquisition.



According to S. 1901 “Cybersecurity Research and Education Act of 2002”

- (A) Cryptography.
- (B) Threat and vulnerability assessment, including risk management.
- (C) Web security.
- (D) Operations of computer emergency response teams.
- (E) Cybersecurity training, education, and management.
- (F) Computer forensics.
- (G) Defensive information

Personal Cybersecurity Tips / Best Practices





cybersecurity is about

❖ availability

❖ Integrity &

❖ secrecy(Confidentiality) of information systems and networks in the face of attacks, accidents and failures with the goal of protecting operations and assets



THE IMPORTANCE OF CYBER SECURITY

As the volume and sophistication of cyber-attacks grow, companies and organizations, especially those that are tasked with safeguarding information relating to national security, health, or financial records, need to take steps to protect their sensitive business and personnel information. As early as March 2013, the nation's top intelligence officials cautioned that cyber-attacks and digital spying are the top threat to national security, eclipsing even terrorism.



Why carry out a cyber security risk assessment?

The process of identifying, analyzing and evaluating risk – is the only way to ensure that the cyber security controls you choose are appropriate to the risks your organization faces. Without a risk assessment to inform your cyber security choices, you could waste time, effort and resources – there is, after all, little point implementing measures to defend against events that are unlikely to occur or won't have much material impact on your organization. Likewise, it's possible that you'll underestimate or overlook risks that could cause significant damage to your organization. This is why so many best-practice frameworks, standards and laws – including the GDPR (General Data Protection Regulation) and DPA (Data Protection Act) 2018 – require them.



Top Five Cyber Risks

When setting up your new business, you are likely to rely on a variety of IT, such as smart devices, PCs and cloud-based systems. You could be holding customer data, employee information and possibly detailed product designs. These are likely to be of interest to cyber criminals – no matter how small your business. An awareness and basic understanding of the threats posed in a cyber-world will help protect your digital assets, intellectual property and your business

Top Five Cyber Risks

1. Ransomware

2. phishing

3. Data Leakage

4. Hacking

5. Insider threat



1. Ransomware

This is a form of malware (malicious software) that attempts to encrypt (scramble) your data and then extort a ransom to release an unlock code. Most ransomware is delivered via malicious emails. Follow these key steps to protect your company.

- Staff awareness:

staff should be wary of unsolicited emails, particularly those that ask for a prompt response.

- Malware protection: install and maintain good antivirus and malware protection software.
- Software updates: keep your applications up to date.
- Data backups: a series of well managed data backups will allow you to recover from an unencrypted version of a file. Make sure you regularly test your backups



2. Phishing

Phishing is an attempt to gain sensitive information while posing as a trustworthy contact, for example a bank or online service. Spear phishing is a highly targeted attempt to gain information from an individual. Phishing emails may look completely convincing, often with faultless wording and genuine logos. There is a form of spear phishing, where a fake email from a CEO applies pressure on a CFO into making an urgent payment, this has become known as Whaling.



2. Phishing

It is worth considering ways to add additional safeguards to protect the identity of CEOs and CFOs to prevent impersonation. Here are a few steps you can use to protect yourself. • Keep in mind that companies simply do not ask for sensitive information. • Be suspicious of unexpected emails. • Make use of anti-malware software. • Make sure you have spam filters turned on. Check them regularly in case they have accidentally trapped an innocent email



3. Data leakage

While cyber security in the office may seem challenging, it is essential to understand that security extends well beyond the office these days. The use of smart phones and tablets has become widespread. The ubiquitous and cheap nature of portable storage devices makes them a useful tool for the backup and transportation of data. Those features mean they are also a target for data thieves. The following pointers provide useful first steps to prevent data leaking from your organization.



3. Data leakage

- Ensure mobile devices have passcode locks.
- Turn on the tracking by GPS and the option to remotely wipe the device if it is lost.
- The use of encryption software is highly recommended when using portable storage devices.
- Keep an eye on your mobile devices and paperwork at all times. A large proportion of crime is opportunistic, taking your eye off your briefcase or smart device could result in a serious loss of data



4. Hacking

Gaining access to IT systems from outside an organization still offers rich pickings for criminals. Traditionally they have attempted to gain access to bank account information or credit card databases. However, intellectual property is another source of value. The use of social engineering, tricking staff into revealing user names and passwords, remains a threat.

The primary methods to protect yourself from hacking are network firewalls, data access security, procedures for providing and removing access, and user awareness and training



5. Insider threat

Insider threat If your organization employs staff (full time or as contractors), there is a possibility they could leak data by mistake or maliciously. The potential damage from a leak of documents cannot be underestimated. Use these tips to mitigate the size of any data leak.



Please remember

In all these areas it is key to remember that alongside technology, well-developed processes, procedures and staff training go a long way to protecting your valuable data. For example, if someone leaves your employment, make sure you remove their access. The reality today is that you should protect your digital assets with the same vigilance as you do when locking your office door at the end of the day.



5. Insider threat

- Educate your team to be alert to issues and minimize careless mistakes.
 - Limit how much data staff has access to. The principle of 'least privilege access' should apply to all IT systems. Only provide staff with the minimum access they need to do their roles.
- Control the use of portable storage devices, such as USB memory keys, portable hard drives and media players.
- Consider using applications in certain situations to monitor staff behavior – who copies what

Agenda

- **State of Cybersecurity/ Current Cyberthreats**
- **Topics of Interest:**
 - **Phishing Emails/Scams**
 - **Ransomware**
 - **Social Engineering**
- **How to protect yourself**
 - **WFH Tips**
 - **Additional resources**
- **Q&A**



Current state of Cybersecurity



Phishing attacks are still number 1



Ransomware attacks are on the rise



Social Engineering is the leading attack vector for scams





Cost of Cybersecurity Attacks

JBS Foods pays \$14m to ransomware attackers

By Ry Crozier
Jun 10 2021
11:23AM

Says the decision to make payment was 'very difficult'.

JBS Foods, the meat processor whose Australian and US operations were hit by a ransomware attack **earlier this month**, paid "the equivalent of US\$11 million" (A\$14 million) to the group behind the attack.

The company made the payment despite the "vast majority of the company's facilities" already having been operationally recovered.



Cyberattack Sees UniSA Systems Shut Down

BY ADMIN ON MAY 18, 2021
APP-ACSM, CYBER SECURITY, EDITOR'S DESK, FEATURED, GOVERNANCE, RISK & COMPLIANCE, HACKING & PENETRATION TESTING, INFORMATION SECURITY, VULNERABILITIES

A cyberattack at the University of South Australia is continuing to impact staff and students. The circumstances surrounding the attack remain unclear, but key systems are still offline two days after the University noticed the first outage.

"UniSA experienced a cyberattack on the weekend which caused an outage of its staff email system," a UniSA spokesperson told MySecurity Media. "The University is still investigating the issue."



Targeting scams report 2020

Losses

\$851 million

2020 combined financial losses to scams as reported to Scamwatch, ReportCyber (ACSC), ASIC, other government agencies and 10 financial institutions (ANZ, Commonwealth Bank, NAB, Westpac, BoQ, Bendigo and Adelaide Bank, Macquarie Bank, Suncorp, Western Union and MoneyGram)

\$176 million

Amount reported lost to Scamwatch

216,087

reports to Scamwatch



Nine Network under attack by cyber hackers, threatening news services nationwide



By 9News Staff | 7:04am Mar 29, 2021



Tweet

POLITICS

Colonial Pipeline paid \$5 million ransom one day after cyberattack, CEO tells Senate

PUBLISHED TUE, JUN 8 2021:10:17 AM EDT | UPDATED WED, JUN 9 2021:8:24 AM EDT

NEW ZEALAND / MEDIA & TECHNOLOGY

Waikato DHB cyber attack: Medical files may have been taken

5:46 pm on 24 May 2021

Share this     

NSW driver's licence data stolen in Accellion breach

By Justin Hendry
Feb 25 2022
6:30AM

Some customers, agency staff only now being notified.

Driver's licence details were among the personal information stolen from Transport for NSW in the Accellion data breach last year, *iNews* can reveal.



Phishing Attacks



Phishing emails
e.g. COVID 19 vaccines

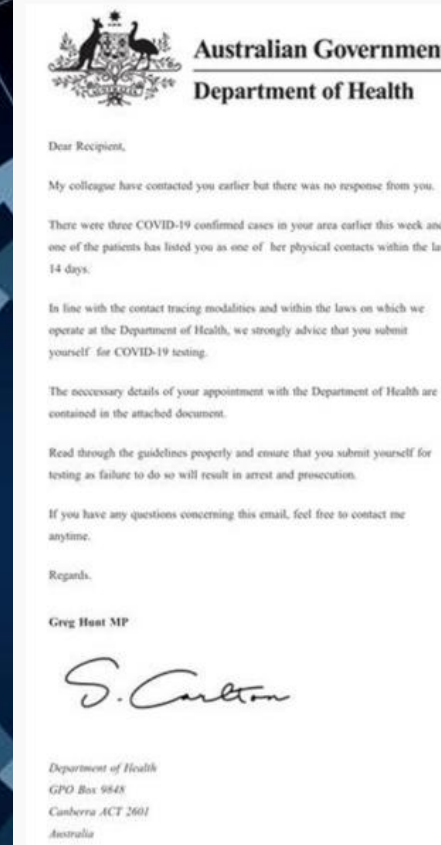


Vishing
e.g. Computer technician call to
fix a virus on your machine

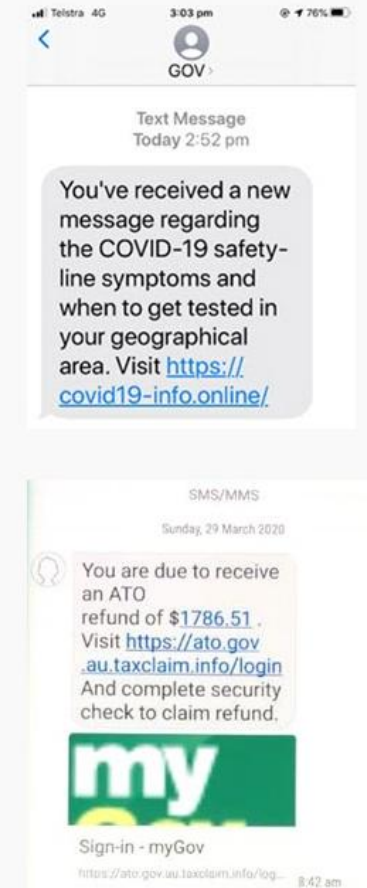


Smishing
e.g. ATO text messages to claim a
tax refund

Department of Health impersonation email



Fake myGov texts



Phishing Examples

[EXT] NOTICE:Disciplinary Action-Oral Warning - Temporary Items

Message

Delete Reply Reply to All Forward Attachment Meeting Switch Background Move Junk Rules Read/Unread Categorise Follow Up

1. Subject line of email is vague and evokes urgency.

[EXT] NOTICE:Disciplinary Action-Oral Warning

Vice-Chancellor <vc@unimelb.edu.au>

The University of Me... 177.6 KB

Download All Preview All

2. Email has been tagged as an external source confirming it is spoofed.

Today at 12:22 pm

UoM notice: External email. Be cautious of links, attachments, or impersonation attempts

NOTICE:Disciplinary Action-Oral Warning

This letter serves as documentation of an oral warning, delivered on April 12, 2021, concerning your behaviour with a senior colleague. The university of Melbourne has made the decision to take disciplinary action regarding the complaint filed against you by the staff.

Again this letter serves as documentation of company name oral warning concerning your hostile aggressive behaviour. A photo of the complaint form is attached with the email.

Sincerely

3. Email content is vague and uses fear as "bait" for the user to click/download on the attachment.

4. The UoM logo is placed before the senders signature.



THE UNIVERSITY OF
MELBOURNE

The vice-chancellor
Prof Duncan Maskell
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The University of Melbourne
Victoria 3010 Australia
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Phishing Examples

From: courrier <r07fra@tempmail.top>

Reply to: "r07fra@tempmail.top" <r07fra@tempmail.top>

Date: Wednesday, 28 April 2021 at 10:39 am

To: [REDACTED]

Subject: Your Package #4687890568 is ready for delivery.

1. Spelling error in
senders address

2. Email subject
line is vague

Open in your [web browser](#)

Your Package #4687890568 is ready for delivery.

Failed delivery attempt: 28/04/2021

Your parcel was returned to our depot and you need to reschedule your
package delivery.

To receive your package, we ask that you send us your correct address and
pay the new shipping costs "1.99\$" at the following link:

[COMPLETE MY DELIVERY ADDRESS](#)

Thank you,

3. Requesting financial
and personal
information

4. Link leading to an
external website

5. Signature is
missing, obscure

Sent by GlobalCourier

Chris

If you wish to unsubscribe, please [click here](#).



Ransomware

Type of malware that encrypts data specifically asking for payment in order to restore access.

How?



Email attachments



Website downloads



Email links



Website links

Protect Yourself



Regular backups



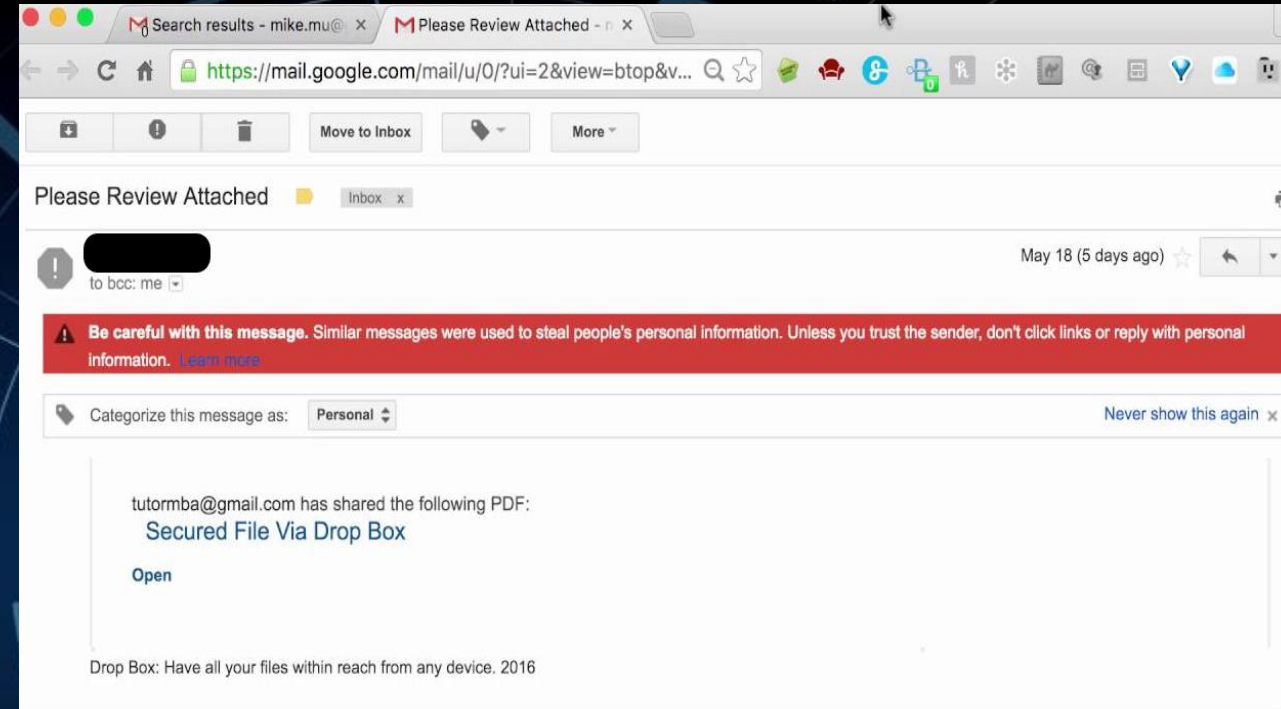
Updates



Verify emails



Don't PAY!



Why is Cyber awareness so important?

- Everything is CONNECTED!
- Personal documents
- Identity
- Finances
- Digital footprint



How can I protect myself?



Passwords

- Long and strong. Passphrases
- Enable 2FA where possible
- Change default passwords
- Don't reuse passwords across accounts
- **Use a Password manager (LastPass is FREE)**

LastPass ****



';--have i been pwned?

Check if you have an account that has been compromised in a data breach

email address

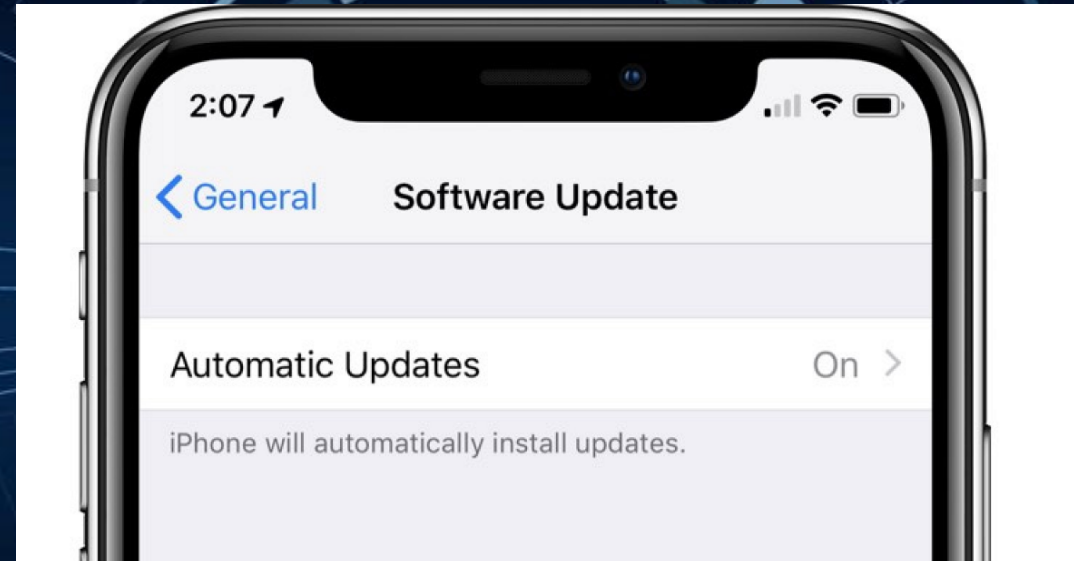
pwned?



**STRONG
PASSWORDS.**

Updates

- Ensure all devices are on their latest updates.
- Turn on AUTOMATIC UPDATES
- Make time for updates
- Spring clean your apps regularly



Be aware of Scams

1. Phishing - email
2. Vishing – phone call
3. Smishing – text messages

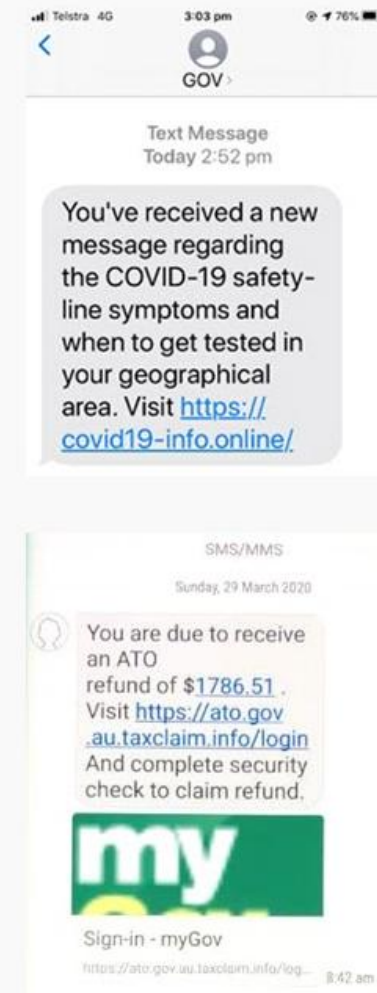
Look out for:

1. Urgency
2. Asking for personal/financial information
3. Unsolicited
4. Contain links and downloadable files
5. Bad grammar
6. Too good to be true

Department of Health impersonation email



Fake myGov texts



Search Yourself (Digital Footprint)

- Privacy controls
- Be mindful of what you share
- Review app privacy collection



Creating a Cyber secure home checklist

- Are my devices secure?
- Using VPN to access University systems?
- Beware of using FREE Wi-Fi
- Do I have anti virus installed?
- Am I backing up my important files? Cloud & Local
- Are my devices up to date?
- Enable two-factor authentication (2FA) where possible
- STOP. THINK BEFORE YOU CLICK.





Helpful Websites

- Scamwatch
- Stay Smart Online
- SANS Security Awareness Blog

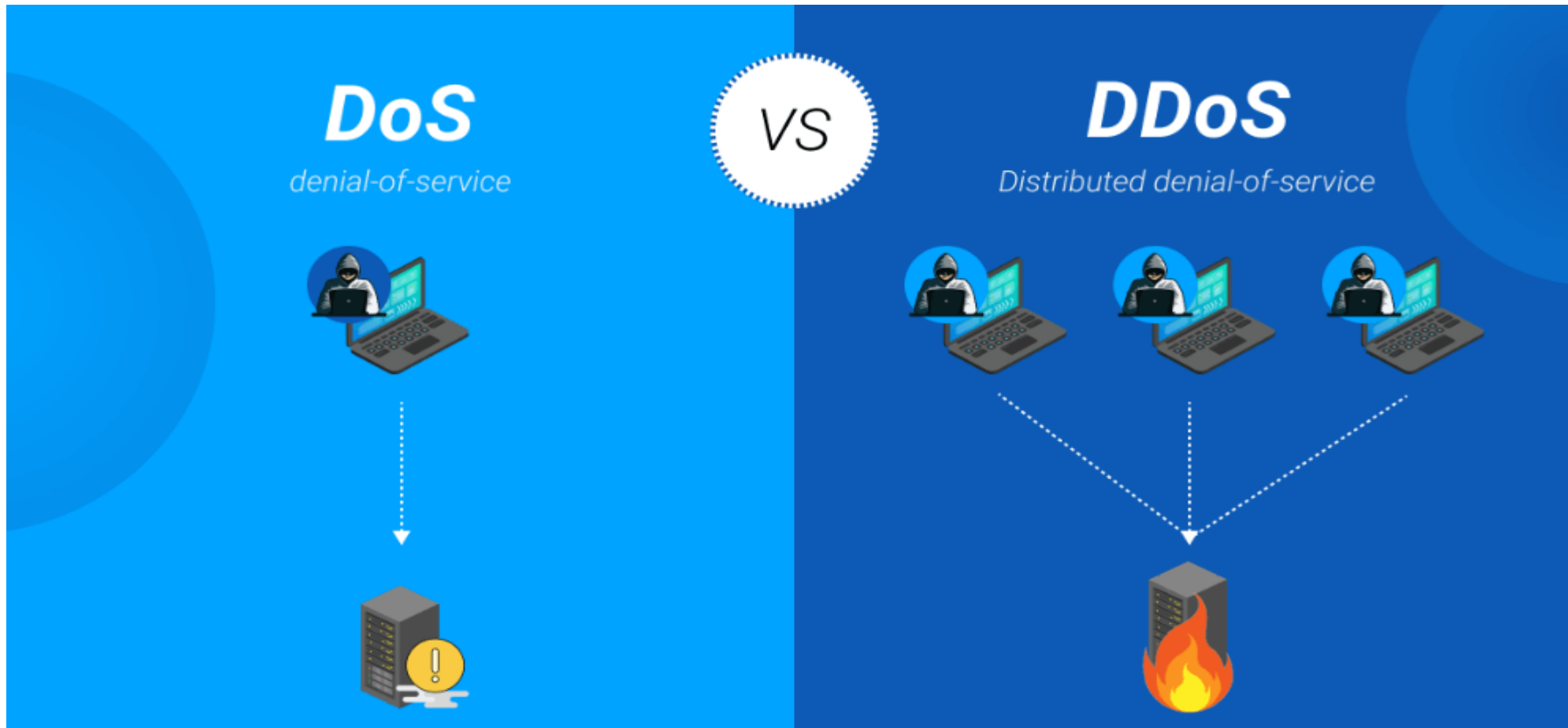


SANS Security Awareness Blog

Read the latest in security awareness, walk away with actionable tips and insight in keeping your organization secure.

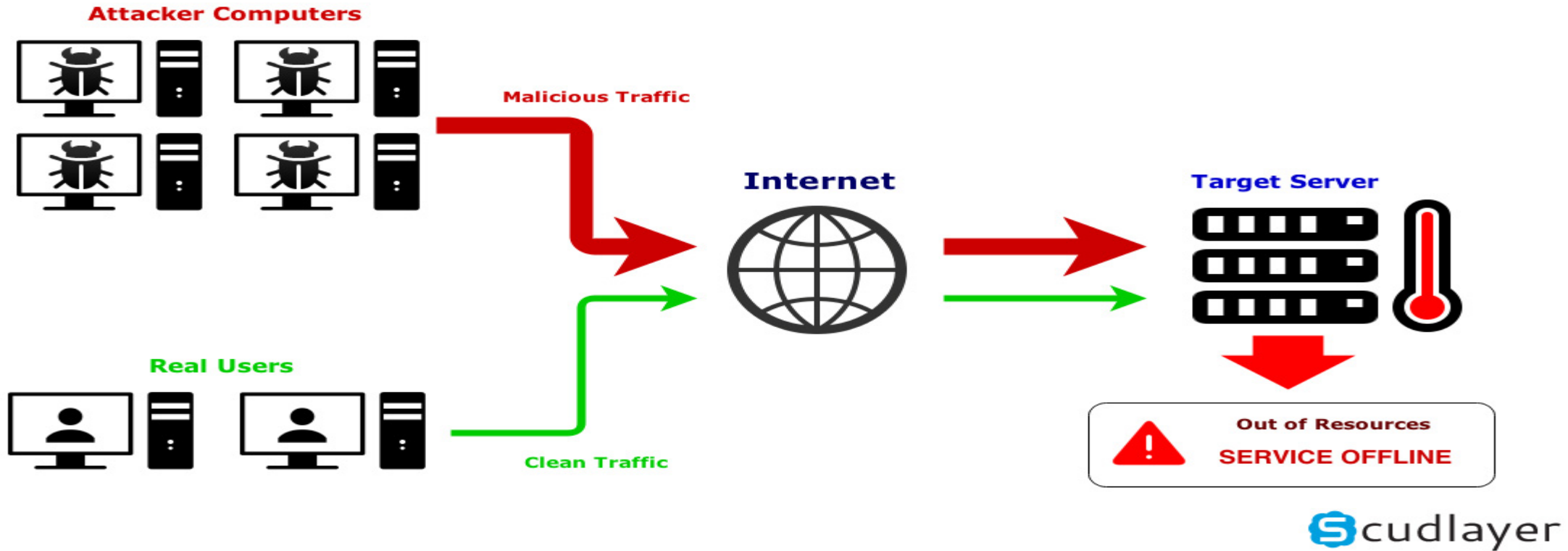
DOS and DDOS Attacks?

How we can forget DOS and DDOS attack? Are the most commonly used type of attack on systems connected via internet specially designed for financial ,ecommerce and government purposes.



DDOS ATTACK

Operation of a DDoS attack





Any Question?
Let have discussion